

Concatenative Synthesis

Granular vs. Concatenative

Granular Synthesis is a classical synthesis technique whose main idea is that of generating sound from a number of short sonic excerpts with certain timbral characteristics.

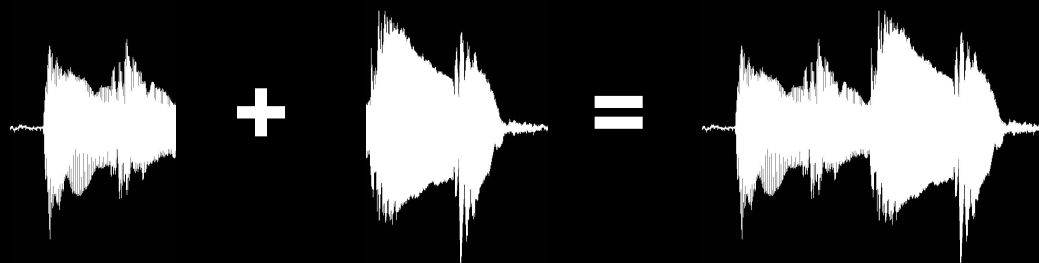
Concatenative Synthesis is a synthesis technique largely based on Granular Synthesis that uses some sort of analysis to classify and select the output grains.

Concatenative Synthesis

“Creative” and “artistic” application of MIR

Create a new sound (**or rather reconstruct a target sound**) by stitching existing audio snippets together, according to some rules and procedures...

(Kind of) automatic and continuistic “micromontage” based on audio descriptors



Inspirations

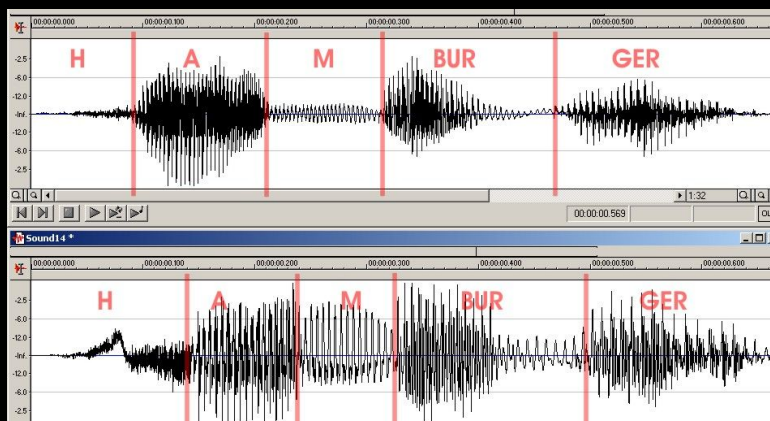
Plunderphonics (1985 John Oswald)

Mashups of existing audio recordings

Challenge to notions of authorship, piracy, etc.

Digital Speech Synthesis (Hunt and Black 1996)

Unit selection of speech phonemes

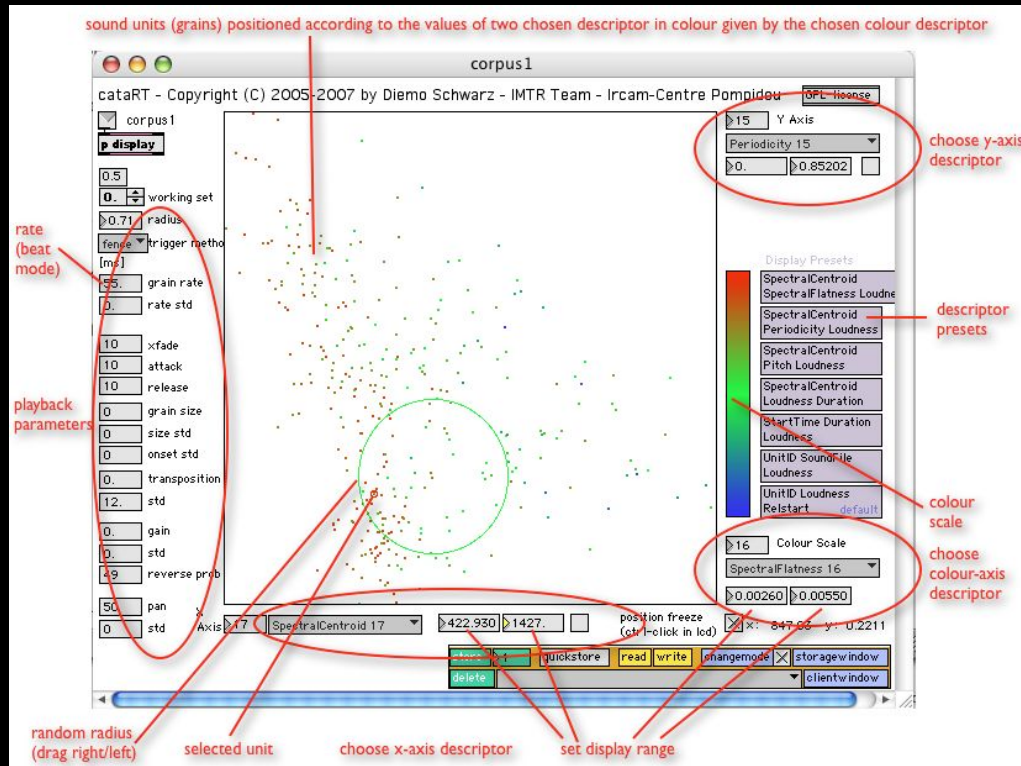


Concatenative Synthesis Steps

Given a (bank of) soundfile(s), do this:

1. **Onset detection** (find where events start)
2. **Segmentation** (split file(s) into short “units”)
3. **Feature extraction** typically
 - a. Some timbral descriptor such as MFCC
 - b. Spectral energy indicators (noisiness, centroid, etc...)
4. **Database creation** with extracted features
5. **Target sound analysis**
6. **Concatenation** of units to reconstruct original sound based on some similarity metric (NN, K-means, HHM)

Concatenative Synthesis Systems



CatArt (Schwarz 2006)

Max/MSP

AudioGarden (Frisson 2010)

EarGram (Bernardes, 2012)

Pd

RhythmCat (O’Nuanain, 2016)

JUCE, Essentia, Vst

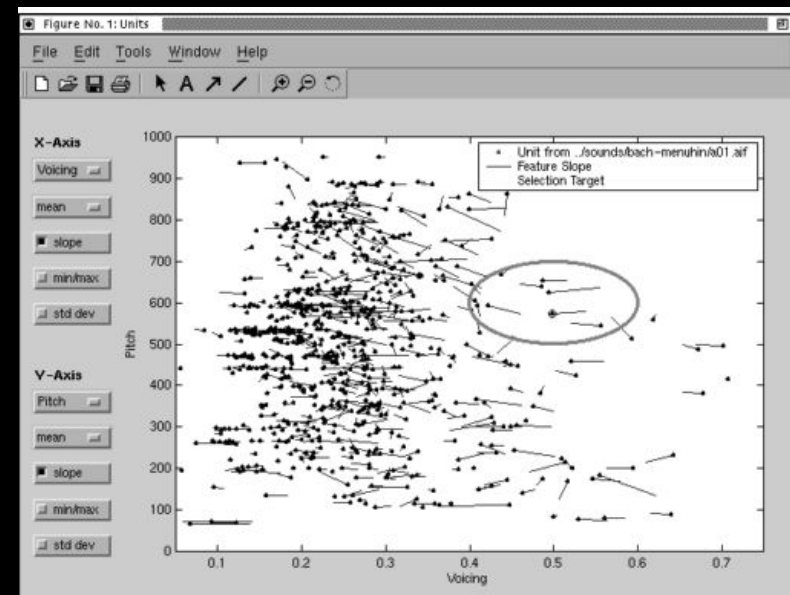
Early Digital Musical Concatenative Synthesis

Caterpillar (Schwarz, 2000)

- Unit Selection using Viterbi Algorithm (HMMs)

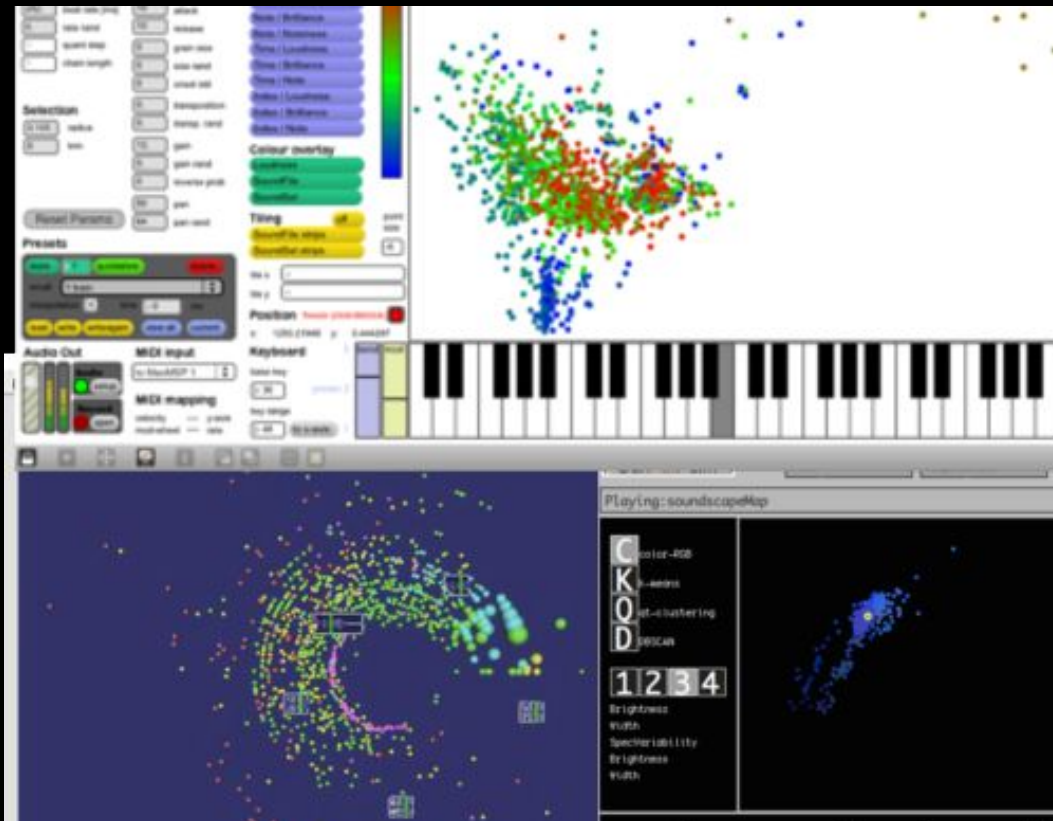
- **Musaicing** (Zils & Pachet 2001)

- Musical Mosaicing
- Constraint Satisfaction
 - “Mimic” Mode to model a target
 - User selectable constraints



State of The Art Systems

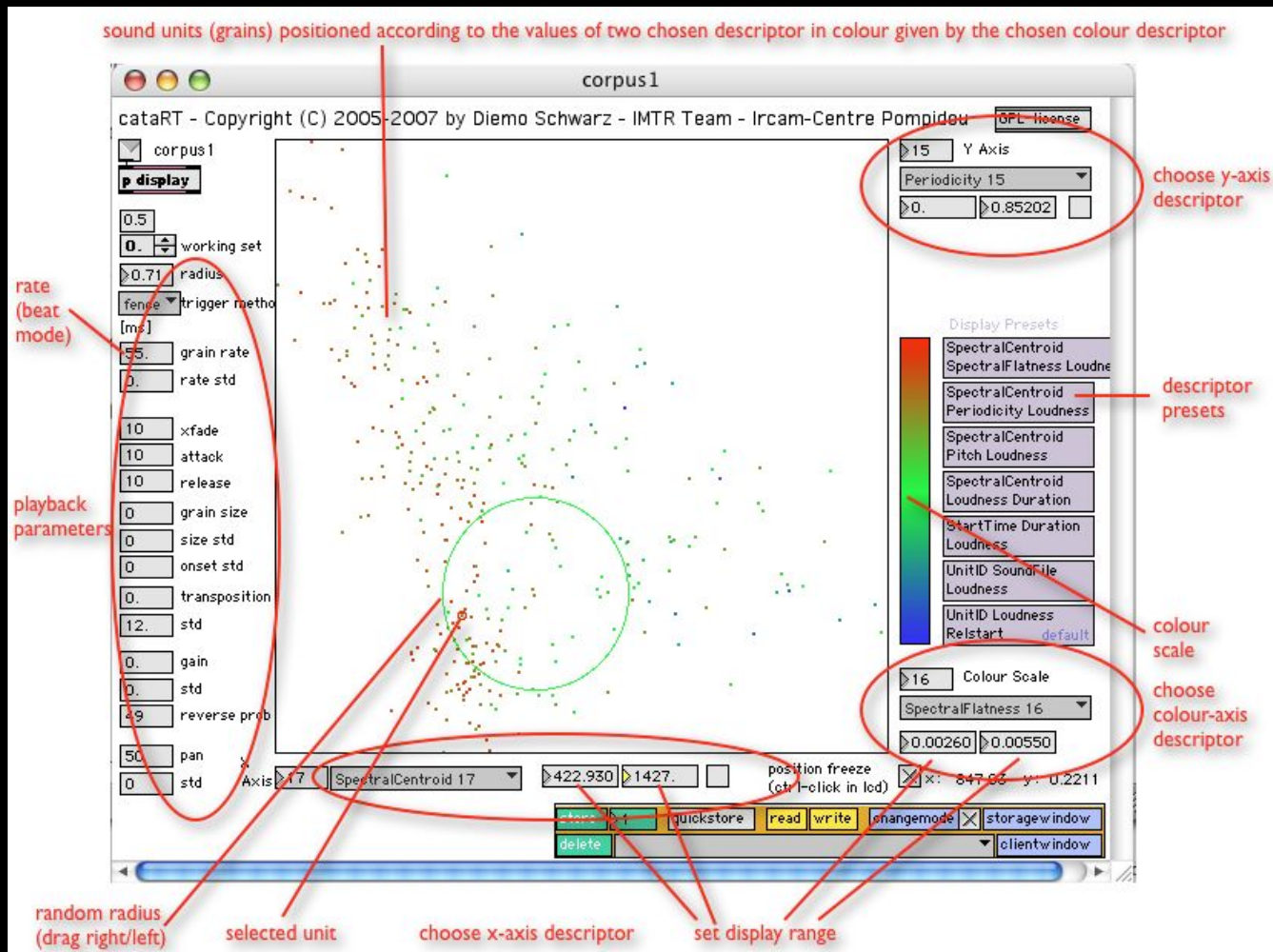
CatArT (Schwarz, 2006)



AudioGarden (Frisson et al., 2010)

EarGram (Bernardes, 2014)

CataRT - https://www.youtube.com/watch?v=cWXdTlu_n44

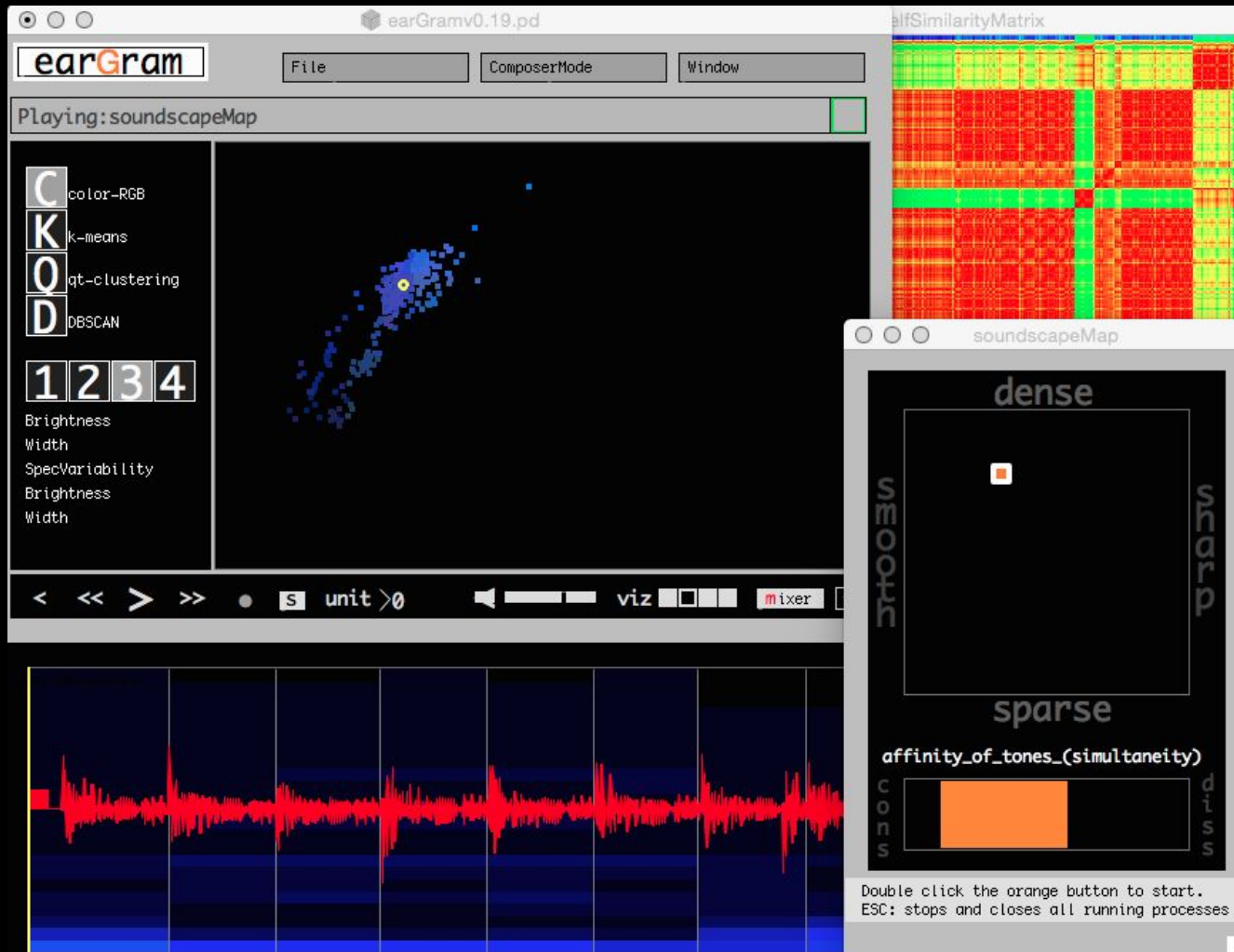


CataRT

- Real-time concatenative Synthesis
 - IRCAM (Schwarz 2006)
 - Max/MSP & FTM (data structures)
- 2D Timbre Space
 - Select features that appear on each axis
- Unit Selection
 - Proximity of of sound units in timbre space to the mouse
 - Complex unit selection procedures are not as conducive to real-time performance!

Schwarz, D., Beller, G., Verbrugghe, B., & Britton, S. (2006). Real-Time Corpus-Based Concatenative Synthesis with CataRT. *Proceedings of the 9th International Conference on Digital Audio Effects*, 18–21. Retrieved from <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.104.9605&rep=rep1&type=pdf>

EarGram - <https://vimeo.com/80322437>



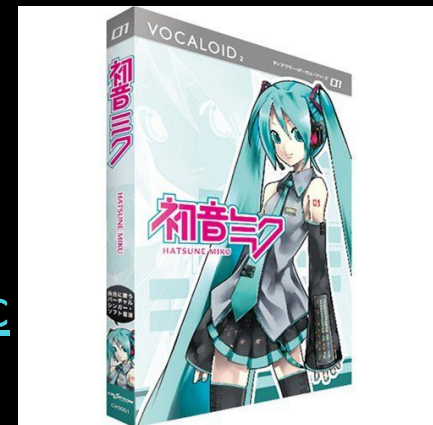
EarGram

- Real-time concatenative Synthesis
 - Porto SMC (Bernardes et al, 2013).
 - Pure Data!
- 2D Timbre Space
- Some other nice features
 - Self-similarity matrix
 - 2D Soundscape Control
 - Tempo support

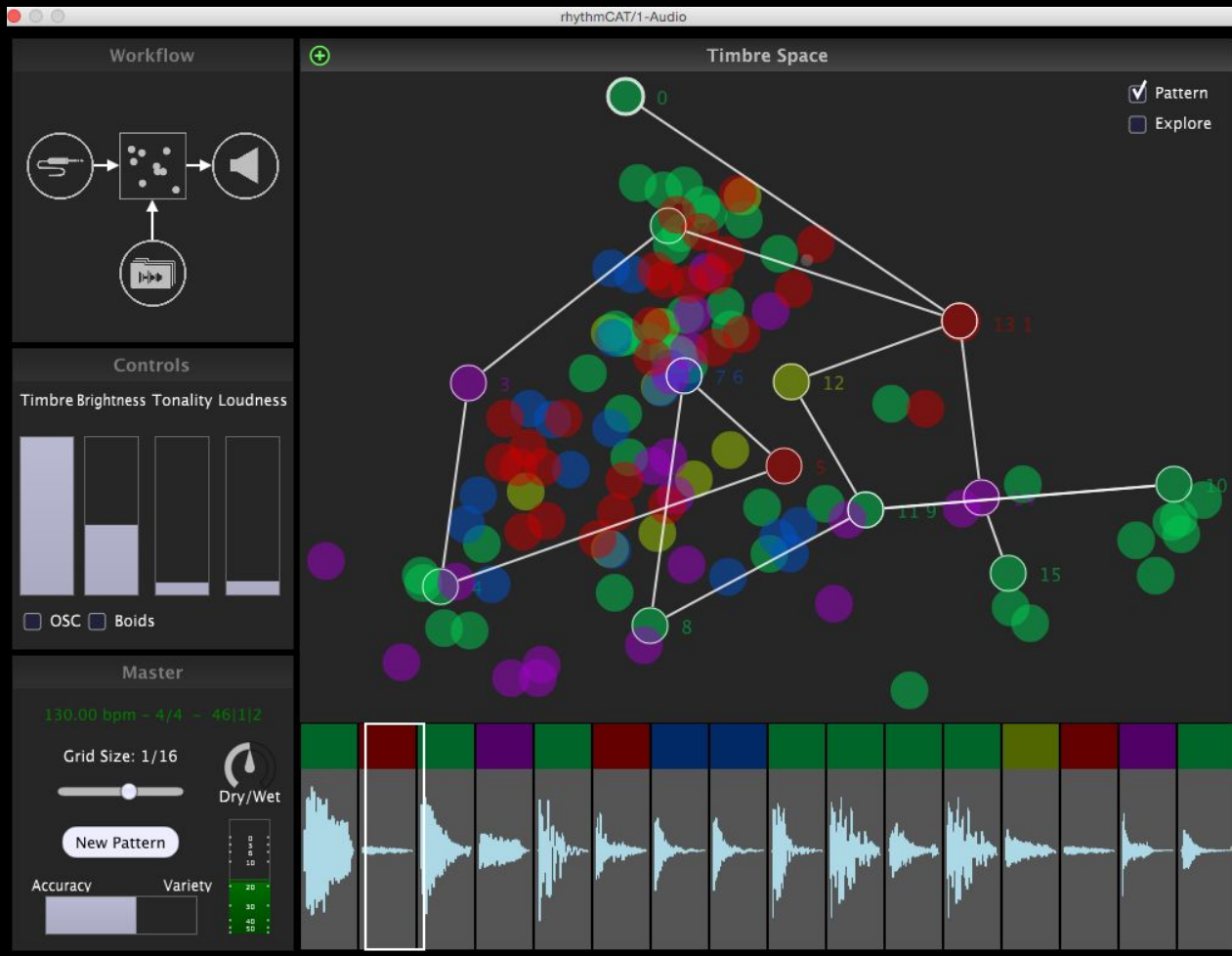
Bernardes, G., Guedes, C., & Pennycook, B. (2013). EarGram : An Application for Interactive Exploration of Concatenative Sound Synthesis in Pure Data. *From Sounds to Music and Emotions*, 110–129. Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-41248-6_7

Commercial Systems

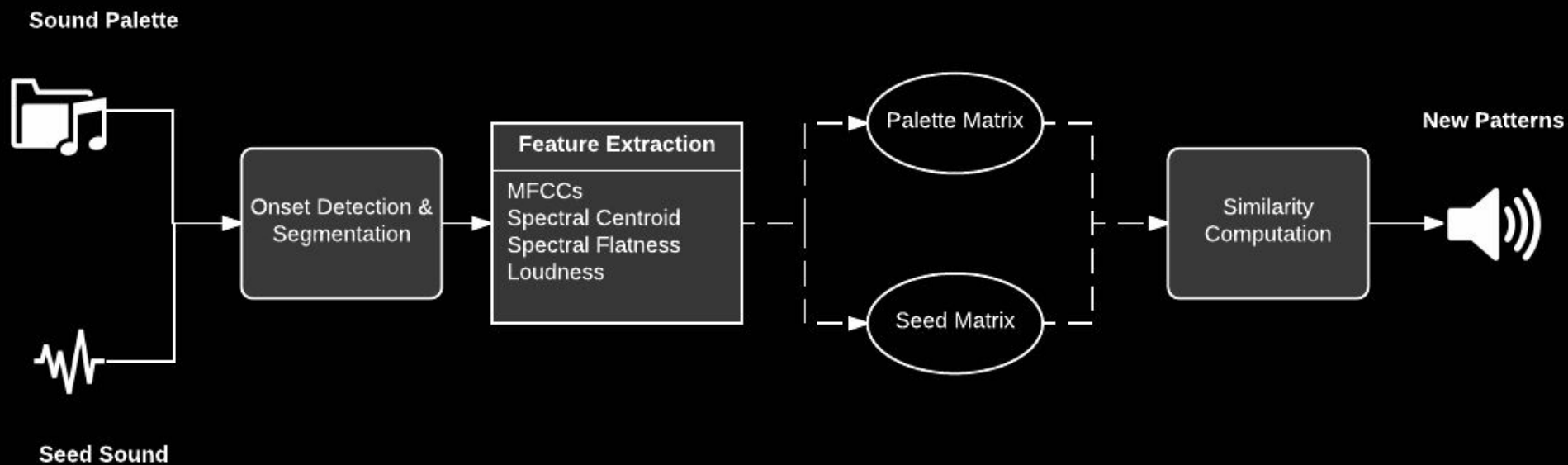
- LoopMash
 - Steinberg + MTG
 - VST/iOS
 - Waveform centric
 - http://www.steinberg.net/en/products/mobile_apps/loopmash.html
- Vocaloid
 - Yamaha + MTG
 - Singing synthesis
 - <https://www.youtube.com/watch?v=FoTd918zhZc>
 - Cultural phenomenon in Japan



RhythmCAT



RhythmCAT Implementation

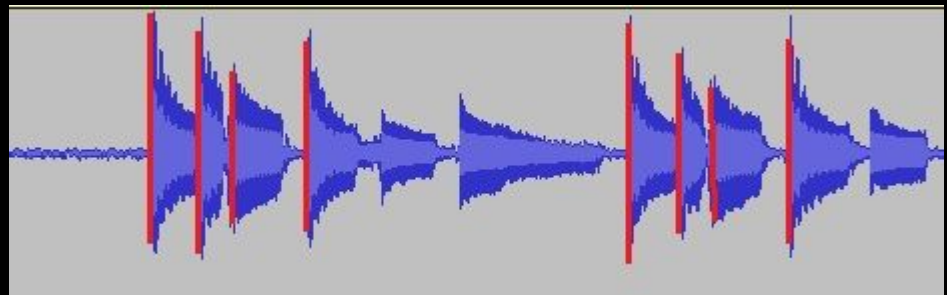
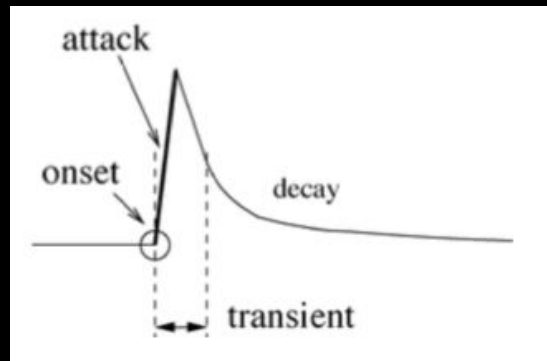


RhythmCAT

- Rhythmic Concatenative Synthesis
 - Mimic a rhythmic loop of audio using completely different audio
 - VST plugin
 - 2D Timbre Exploration with Graph-based manipulation
 - User-focused
- Publications:
 - Ó Nuanáin, C., Jordà, S., & Herrera, P. (2016). An Interactive Software Instrument for Real-time Rhythmic Concatenative Synthesis. In *New Interfaces for Musical Expression*. conference, Brisbane, Australia.
 - Ó Nuanáin, C., Jordà, S., & Herrera, P. (2016). Towards User-Tailored Creative Applications of Concatenative Synthesis in Electronic Dance Music. In *MUME 2016 - The Fourth International Workshop on Musical Metacreation*. Paris.
 - Ó Nuanáin, C., Herrera, P., & Jordà, S. (2016). An Evaluation Framework and Case Study for Rhythmic Concatenative Synthesis. *Proc. 17th International Society for Music Information Retrieval Conference*.

Onset Detection -> Segmentation

- Many approaches exist for detecting start of notes in signals
 - (Bello et al. 2005)
- Typical stages include:
 - Reduce signal to onset detection *function* e.g. estimating changes in spectrum from frame to frame
 - Perform peak peaking to find local maxima corresponding to onsets



Features

- MFCCs
 - Used in speech recognition
 - Very useful in comparing the *timbre* of two different sounds
- Spectral Centroid
 - Centre of “Gravity” of energy in spectrum
 - Useful for differentiating low freq sounds from high freq (kick vs snare).
- Spectral Flatness
 - Flat spectra are related to more noise-like sounds (hi-hats)
- Loudness

Similarity Computation

- Compute weighted Euclidean Distance between unit feature vectors

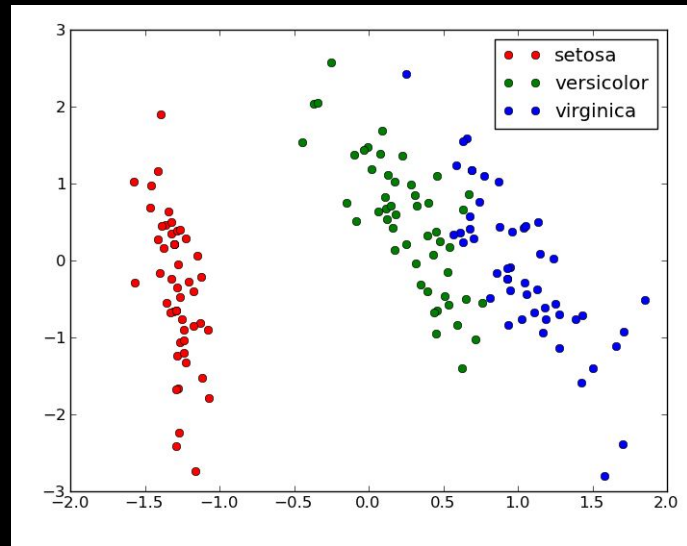
$$C_{i,j} = \sqrt{\sum_{k=1}^n w_k (a_k - b_k)^2}$$

- N*M ascendingly sorted matrix of unit indices
 - N is size of palette
 - M is grid length (8, 16, 32 etc.)

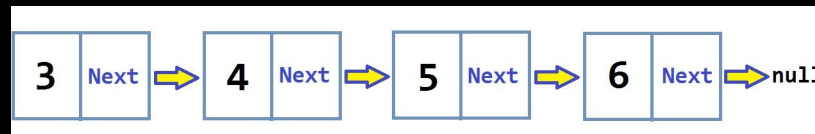
5	0	3	7	5	4	6	4	4	6
4	6	1	0	7	6	3	5	3	9
1	5	1	8	6	6	5	0	7	3
7	4	8	4	7	6	6	5	6	1
0	0	7	6	0	2	9	7	5	0
6	2	6	9	4	6	6	6	8	6
1	0	2	5	4	6	7	4	0	2
7	0	4	6	7	3	2	7	7	7
1	9	3	0	3	2	8	9	2	2
9	8	0	0	1	1	4	0	6	1

Visualisation and Interaction

- Weighted feature vector -> PCA -> Scaled 2D Coordinates



- Linked List represents connections between units in sequence efficiently.



Technologies

- **JUCE**
 - C++ framework particularly suited for audio applications
 - OpenGL Acceleration
- **Essentia**
 - In-house, open source library for audio and musical analysis
 - <http://essentia.upf.edu/>
- **OpenCV**
 - Matrices
 - PCA
 - Fast mathematical operations (sorting, distance computations).

TimbreID: A Pd library for feature extraction

Easy to use and Pd-idiomatic library with many spectral and time based descriptors

Real-Time and NON-Realtime modes (useful for live input and pre-recorded sources!)

Simple aggregation and accumulation inside PD

Basic clustering, normalisation and classification tools included in [timbreID] object

DEMO: RhythmCAT, earGram and timbreID Library